SIMATS ENGINEERING

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Course Code: **CSA0820** **Slot: C**

Course Name: PYTHON PROGRAMMING FOR SCIENTIFIC COMPUTING

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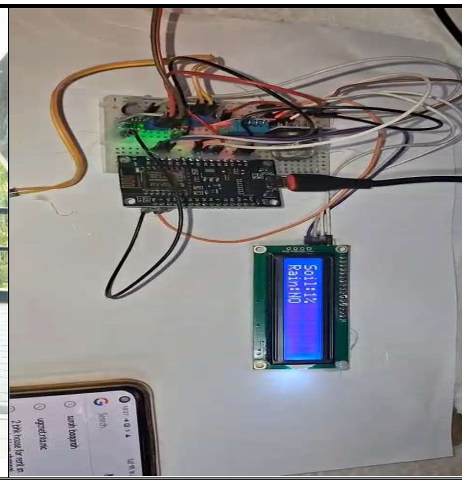
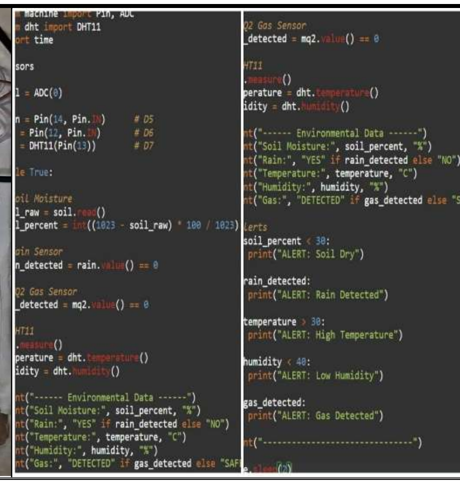
Project Title: IoT Based Smart Environmental Monitoring System

Module Photographs:





Project Description:

An **IoT Based Smart Environmental Monitoring System** is an intelligent system that uses sensors and Internet of Things (IoT) technology to monitor environmental conditions in real time. The system collects data such as temperature, humidity, air quality, soil moisture, rainfall, and gas levels using various sensors connected to a microcontroller like Node MCU or ESP8266. This data helps users understand the current environmental status and make informed decisions.

The collected sensor data is processed by the microcontroller and transmitted to a cloud platform through a Wi-Fi connection. Users can access the data remotely using a web dashboard or mobile application. The system continuously monitors environmental parameters and can generate alerts when abnormal conditions, such as high gas concentration or low soil moisture, are detected. This allows quick action to prevent environmental and safety issues.

This system has applications in smart agriculture, weather monitoring, industrial safety, pollution control, and smart city development. It reduces the need for manual monitoring, improves accuracy, and provides real-time information from any location. By integrating sensors, communication networks, and cloud services, the IoT-based environmental monitoring system offers an efficient and cost-effective solution for environmental management and protection.

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